Advanced Lab Session: Non-Linear Dimension Reduction & Latent Variables

PCA, t-SNE, UMAP, and Autoencoders

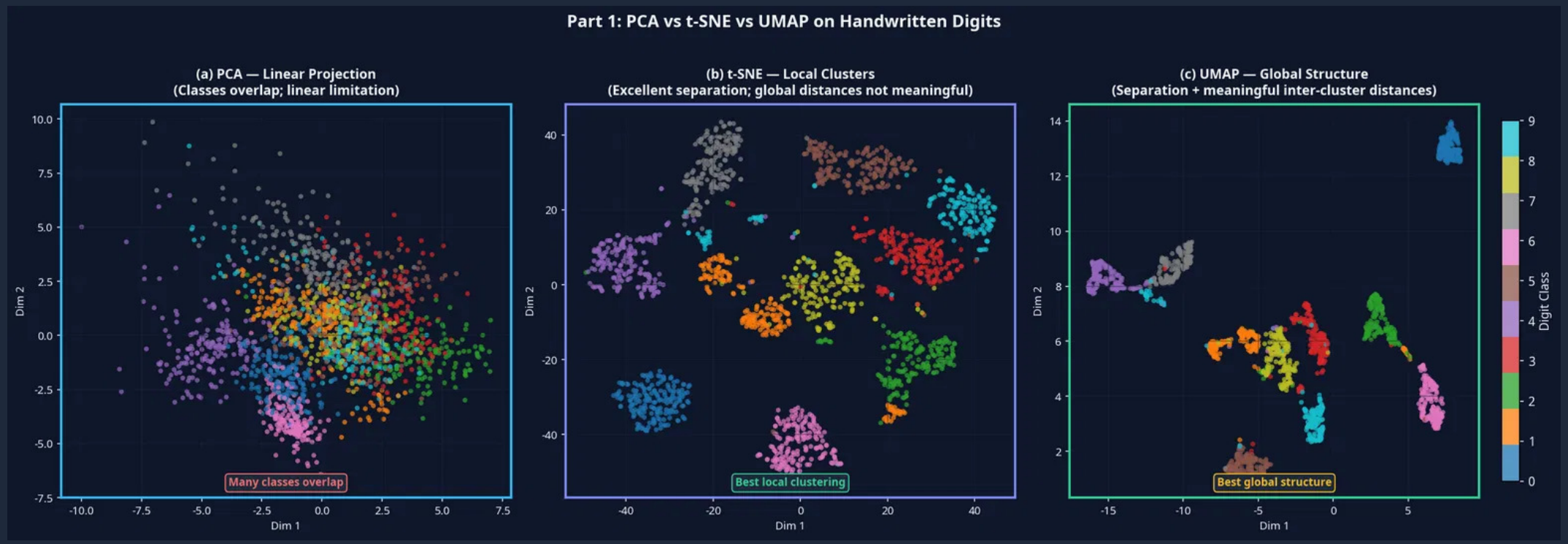
INTERACTIVE PYTHON DEMONSTRATIONS

Lab Overview and Goals

In this advanced interactive Python lab session, we will explore powerful non-linear techniques that extend beyond Principal Component Analysis (PCA). The session is divided into four main parts, followed by a student coding exercise.

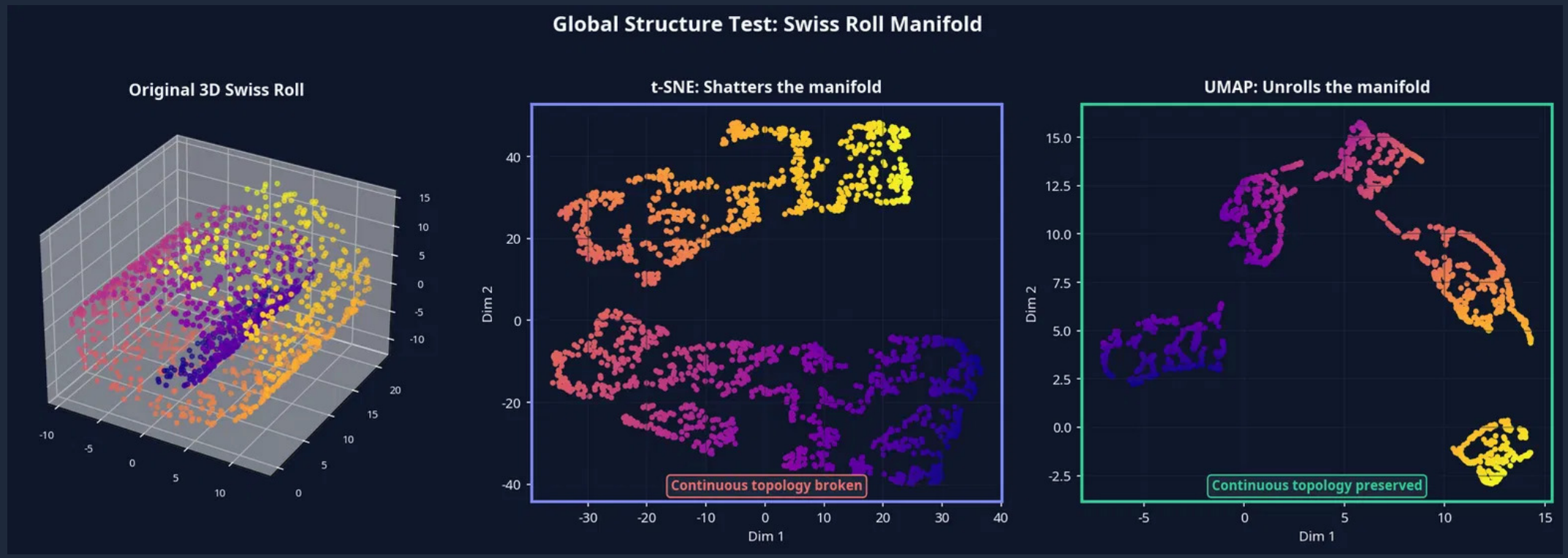
Part 1: PCA vs t-SNE vs UMAP

Comparing three dimension reduction techniques on the Digits dataset. **PCA** (linear) struggles with complex variations, causing overlap. **t-SNE** creates excellent local clusters but loses global relationships. **UMAP** balances both, separating clusters while maintaining meaningful distances between them.



Part 1: Global Structure (Swiss Roll)

To explicitly test global structure preservation, we use the 3D Swiss Roll manifold. **t-SNE** focuses so heavily on local neighborhoods that it shatters the continuous manifold into disconnected clusters. **UMAP**, grounded in topological data analysis, successfully unrolls the manifold while preserving its continuous nature.

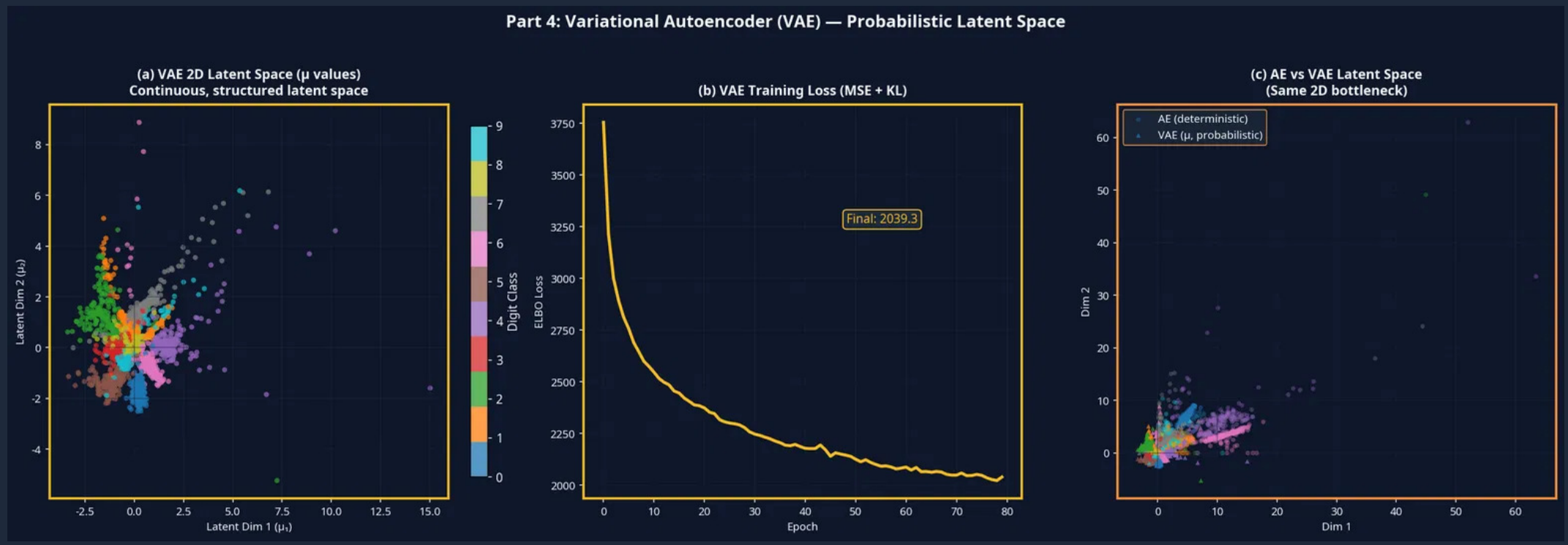


Part 2 & 3: Autoencoder vs Sparse Autoencoder

Standard Autoencoders often learn **dense**, entangled representations where many neurons fire simultaneously. By adding an L1 sparsity penalty to the latent layer, a **Sparse Autoencoder** forces the network to learn disentangled, interpretable features, activating only a few neurons per sample.

Part 4: Variational Autoencoder (VAE)

Unlike standard Autoencoders, a **Variational Autoencoder (VAE)** is a probabilistic generative model. It maps inputs to a distribution (mean and variance) rather than a fixed point. Using the **Reparameterization Trick** and the ELBO loss, it learns a continuous, structured latent space suitable for generating new data.



Method Comparison Summary

A comprehensive summary of the dimension reduction and latent variable techniques covered in this lab. Understanding the trade-offs between linear vs. non-linear, local vs. global preservation, and deterministic vs. generative models is crucial for selecting the right tool for your data.

Method Comparison Summary

	Linear?	Preserves Local	Preserves Global	Generative?	Interpretable Features	Handles New Data
PCA	Yes	Partial	Yes	No	No	Yes
t-SNE	No	Yes	No	No	No	No
UMAP	No	Yes	Yes	No	No	Yes
Autoencoder	No	Partial	Partial	No	No	Yes
Sparse AE	No	Partial	Partial	No	Yes	Yes
VAE	No	Partial	Partial	Yes	No	Yes